RealQM as Plasma Physics

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August 20, 2026

Mathematical formulation by the author; analysis developed with AI assistance (Claude, Anthropic).

Abstract

In this article we extend RealQM — developed for atoms, molecules and nuclei, where matter is modelled as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy — to the plasma physics of **two-component systems**: ions and electrons, both charged and both free to move. A metal is such a system, as are a stellar interior and the recombination era of a Coulomb cosmology; and so, notably, is RealQM itself, which contains discrete charges and no smeared continuum anywhere. The central result is that RealQM's prescription for a metal *coincides* with the Wigner–Seitz construction: one electron domain per ion, filling the cell, with the normal derivative vanishing at the cell face by symmetry. Across the alkali series Li–Cs it reproduces the equilibrium radius to a mean error of 6.2% against the standard treatment's 5.6%, carrying neither an exchange term nor a band-filling term — but that agreement is shown here *not* to be a test of the quantum treatment, since setting the electronic kinetic energy to zero altogether changes the radii by about one per cent. The lattice scale of a simple metal is electrostatic. The bulk modulus does discriminate and is given worse by the framework (79% against 44%). The plasma frequency $\omega_p^2 = 4\pi n$ comes out exactly, since a rigid displacement leaves the localisation energy untouched. Conduction is examined and the framework's answer is the opposite of what a constrained barrier scan suggests. Placed on a ring of hydrogens, an extra electron does not sit anywhere: it relaxes to a uniform corona around the whole ring, at $a = 2.2\,a_0$ and still at $a = 4.0\,a_0$ where a real electron would occupy one atom as H^-. The energy is then flat as the charge is slid round, so there is no corrugation and no resistance — but equally no localised carrier, and the framework never localises charge at any separation. That over-delocalisation is the vanishing interface cost seen from the conducting side: spreading a domain is free, so nothing opposes it and no metal–insulator transition can occur. Barrier scans that impose a localised carrier measure the price of that constraint rather than of transport, which is why they report the opposite. What the framework does *not* have at all is a Fermi surface, and that is the standing open question: de Haas–van Alphen oscillations, the linear electronic specific heat and positive plasmon dispersion are measured in real metals, not in any idealisation, and RealQM has neither bands nor E_F to account for them. A closing remark treats the uniform electron gas, where the localisation term vanishes identically — a result about a construction containing a smeared inert background, which RealQM does not contain, and so a diagnosis of mismatched ontology rather than a test the framework fails.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to the plasma physics of **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system. So is a stellar interior, and so is the recombination era of a Coulomb cosmology.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

A two-component plasma of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 Metals: a Wigner–Seitz problem the framework already agrees with

Take a metal: a lattice of ions, one valence electron each. RealQM assigns each electron its own domain; by translational symmetry those domains are the Wigner–Seitz cells, one ion at the centre of each; and at a cell face, neighbouring cells being identical, the normal derivative of the density vanishes.

That is exactly the **Wigner–Seitz construction**, used for alkali metals since 1933. The framework does not improvise in condensed matter — it arrives already agreeing with an established method, and differs from it in one term only. Standard theory solves the same cell problem for the band bottom and then *adds* the mean kinetic energy of filling the band, $\frac{3}{5}E_F$, together with exchange and correlation, all consequences of the exclusion principle. RealQM keeps the electrostatics, takes its kinetic energy from the cell state itself, and adds nothing: non-overlap is supposed to do that work.

For each alkali metal, with an Ashcroft empty-core pseudopotential of radius r_c, electrostatics $-0.9/r_s + 1.5\,r_c^2/r_s^3$ for a point ion in a neutral sphere, exchange $-0.458/r_s$ and Wigner correlation, minimising the energy per electron over r_s gives Table 1. Each metal contributes one parameter, its core radius, and two measured numbers; across the series the equilibrium radius varies by 70% and the bulk modulus by a factor of six, so tracking the trend is a stronger statement than matching any single case.

		r_s (a_0)			B (GPa)		
	r_c	RealQM	standard	measured	RealQM	standard	measured
Li	1.06	2.40	2.86	3.25	49.6	23.9	11.6
Na	1.67	3.77	3.87	3.93	9.1	8.0	6.3
K	2.14	4.82	4.67	4.86	3.6	4.0	3.1
Rb	2.31	5.20	4.95	5.20	2.7	3.2	2.5
Cs	2.50	5.62	5.28	5.62	2.0	2.5	2.0
mean abs. error		6.2%	5.6%		79%	44%	

Table 1: The alkali series. On the equilibrium radius RealQM tracks the whole trend and is comparable to the standard treatment; on the bulk modulus, a second derivative, both are poor and RealQM is the worse.

The first thing to say about this table is that it does not discriminate. Repeating the calculation with the electronic kinetic energy set to *zero* — valence electrons sharing one territory, for which a uniform density has $\nabla\psi = 0$ exactly — moves the equilibrium radii by about one per cent (Na 3.73 against 3.77, Cs 5.59 against 5.62) and gives a mean error of 7.0%. Pure electrostatics, with no quantum treatment of the electrons at all, reproduces the alkali series about as well as either column above.

That is a property of simple metals rather than a defect of the calculation: the equilibrium volume is set by core repulsion against the Madelung attraction, and the electronic terms are small or cancel. The standard column reaches it by cancelling $\frac{3}{5}E_F$ against exchange; RealQM reaches it with both absent. *The lattice scale of an alkali metal is therefore not a test of the quantum treatment*, and the agreement in Table 1 should not be read as one. What it does show is that nothing in the framework's treatment of the electrons *spoils* a result electrostatics already gives — a weaker claim, and the accurate one.

The bulk modulus does discriminate, being a second derivative, and there the framework is the worse of the two: 79% mean error against 44%. Lithium is the outlier throughout, the empty-core description being known to serve it badly.

The mechanism of the agreement should be stated plainly rather than claimed as a triumph. At $r_s = 3.93$ in sodium the cell state carries $\langle T \rangle = 0.004$ Ha against a band-filling term $\frac{3}{5}E_F = 0.0715$ Ha, so RealQM is *not* reproducing degeneracy pressure. It arrives at a comparable answer because it also omits exchange, -0.117 Ha, a term of opposite sign and comparable size, so that their effects on the position of the minimum largely cancel. That cancellation is the framework's own design claim — that non-overlap does the work exchange does — and across five metals it holds for the radius and fails for the curvature, which is what one would expect of a cancellation that is close but not exact.

3 The plasma frequency, exactly

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{2}$$

free of temperature and of any statistical assumption. RealQM reproduces this exactly, for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on.

The same structure suggests what the framework should do about **Landau damping**. RealQM is not a single fluid: N electrons are N domains, each carrying its own current and drift velocity, which is a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, collisionless damping should therefore emerge for the right reason rather than by an inserted term. This has not been computed.

4 Conduction, where the framework is least settled

Electric conduction is the interaction of the electrons with the ion lattice, and here the position is genuinely uncertain.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$ for every material, making copper behave like a semiconductor. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that carried the metals in §2 is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near

2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $HF + H_2O \rightarrow F^- + H_3O^+$ and $HCl + NH_3 \rightarrow Cl^- + NH_4^+$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

The barrier, which is a static quantity. Transport by hopping is governed by a barrier, and a barrier is an energy difference, which the static solver does compute. Placing an excess electron as a domain of its own at successive positions and relaxing each to convergence traces the surface the carrier moves on — the standard route to a small-polaron barrier.

A linear chain answers the wrong question: four hydrogens in a line give 1.16 Ha = 31.5 eV, but that is the cost of pushing the carrier *through* an ion, and stepping $1\,a_0$ aside makes it marginally worse rather than better, because in a wire there is no route from one interstitial pocket to the next except through an atom. A lattice has one. In a 4×4 square lattice the carrier passes from the centre of one square through the *midpoint of a bond* — flanked by two atoms at half a spacing each, never at zero distance from any — to the centre of the next.
The lattice route is real: passing between two atoms costs 5.8 times less than passing through one, so the chain's figure was an artefact of dimensionality. But the barrier that remains is 5.5 eV, some

lattice spacing	pocket (Ha)	saddle (Ha)	barrier
$a = 2.5\,a_0$	-16.254	-16.053	0.200 ± 0.003 Ha $= 5.46$ eV
$a = 1.4\,a_0$	-11.824	-11.402	0.421 ± 0.015 Ha $= 11.5$ eV

Table 2: Barrier for a carrier moving between interstitial sites of a hydrogen lattice. Uncertainties are from the mirror-symmetry check, the two bond midpoints either side of a pocket being equivalent by construction.

$200\,k_BT$, and the carrier is confined to its pocket. At $a = 2.5\,a_0$ that may be right — nearly twice the H_2 bond length, where stretched hydrogen is a Mott insulator in nature too.

The density trend, and why the barrier scans mislead. Compressing the lattice did not close the barrier in those scans; it doubled it. But the scans share a defect that only became clear when the question was posed differently: they impose a *localised* carrier and measure what it costs to move. If the framework has no localised carrier, that cost is the price of maintaining an artificial constraint, not of transport.

What the framework actually does with an extra electron. The clean test is a ring, which has no ends and in which every site is equivalent by construction, so the energy must repeat with the lattice period — a validity check the open-chain scans lacked. Six hydrogens on a ring, one extra unit of charge in the corona outside them, slid round by angle:

arc spacing	over an atom	quarter way	between two atoms
$a = 2.2\,a_0$	-3.51	-3.51	-3.51
$a = 4.0\,a_0$	-3.30	—	-3.30

The energy is flat to the resolution available, and the reason is visible in the solution rather than inferred: the extra charge does not sit anywhere. It relaxes to a **uniform corona around the whole ring**, indifferent to where it was placed. There is no localised carrier to move, so there is nothing to corrugate the path and no resistance to measure.

That holds at $a = 4.0\,a_0$, nearly three times the H_2 bond length, where a real extra electron would occupy a single atom as H^-. *The framework therefore over-delocalises: it spreads charge at every separation tested and never localises it.*

One defect, seen from both sides. This is the same missing confinement cost as before, approached from the conducting side. With free interfaces, spreading a domain is free — that is $C = 0$ — so nothing opposes delocalisation, nothing produces a Mott transition, and the framework conducts at all densities. The barrier scans reported the opposite conclusion because they forced the carrier into a configuration the theory does not adopt; the ring reports what it does.

The consequences line up under one statement. A vanishing localisation term at the interface means: no Fermi pressure in a uniform gas; no screening; a dispersionless plasmon; no bandwidth in the sense that would allow a metal–insulator transition; and charge that delocalises at any spacing. Restoring a finite cost at the interface — the Robin condition of §5, with $\beta a = 1.146$ reproducing

Thomas–Fermi — would supply all of them at once, and in particular would let a carrier localise when the atoms are far enough apart, which is what matter does.

A repair that is not available, and one that is. An obvious response to the barrier is to let valence electrons share a territory, which removes it by construction. The scaling argument above rules that out: sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude. *The partition must be kept.*

What may be relaxed instead is the requirement that each domain carry *exactly* one unit of charge. Suppose a domain may hold two — the excess electron is then not a new region competing for space, but additional occupation of a region that already exists. Two things follow immediately, and neither depends on any coefficient. First, by translational symmetry the energy is the *same* wherever the doubly-occupied site sits, so there is no barrier between sites at all. Second, electrostatics favours spreading the excess: with the interfaces free, the kinetic energy is unchanged at any occupation, so only the Coulomb terms act, and they are lowered monotonically as the excess is shared out. A localised double occupancy is therefore not merely mobile but unstable against delocalising.

The cost of forming it is the on-site repulsion, of order 10 eV for hydrogen-like sites — the Hubbard U, here *computed* from the geometry rather than fitted.

What the framework then is. Non-overlapping sites, occupation 0, 1 or 2, an on-site repulsion U, and transfer between neighbours: that is the **Hubbard model**. Which locates RealQM exactly. As formulated — one unit of charge per domain, always — it is the Hubbard model in the limit $U \to \infty$, where double occupancy is forbidden outright and every site is singly occupied. That limit is a Mott insulator at every density, which is precisely what Table 2 found by direct computation, and it explains the density trend as well: no finite U can be beaten by a bandwidth that does not exist.

The repair is correspondingly minimal. Keep the partition, keep the geometry, keep the Coulomb energetics, and allow occupation to take the value two at a computed cost. The framework then sits where the standard treatment of strongly correlated matter sits — the Hubbard model is the working description of Mott insulators, of the cuprates, and of the materials for which band theory fails — with the difference that its U and its lattice are derived rather than supplied.

We do not carry this out here. What is established is where the obstruction lies: not in the interface condition, not in the geometry, but in the rule of one unit of charge per domain. That rule is also the source of one of the framework's results, charge quantization as a consequence of indivisibility rather than an assumption. The tension is worth stating plainly: *the postulate that explains why charge comes in units is the postulate that forbids a metal.* Matter has it both ways, quantizing the particles while leaving the amplitude on each site continuous, and a partition of real densities has no evident means to do the same.

It is fair to ask whether all conduction might be hopping, band transport being a formalism rather than an observable. The idea is not idle: hopping is the correct description of amorphous and organic semiconductors, of doped semiconductors and polaronic oxides, and in "bad metals" at high temperature the mean free path falls to the lattice spacing and the band picture fails there too. What settles it is the *low*-temperature behaviour of a pure metal. Hopping is thermally activated, so its conductivity vanishes as $T \to 0$; the resistivity of pure copper instead falls from $\approx 1.7\,\mu\Omega\,\mathrm{cm}$ at 300 K to below $0.002\,\mu\Omega\,\mathrm{cm}$ at 4 K, a ratio above 10^3 in good samples, with a mean free path reaching millimetres — 10^7 lattice spacings, which is not a sequence of jumps between neighbours. A hopping framework predicts that copper becomes an insulator on cooling, and the measurement runs the other way by three orders of magnitude. The Wiedemann–Franz ratio, fixed at $L_0 = \pi^2 k_B^2/3e^2$ by carriers that convey heat and charge alike, is a second such constraint.

The regimes are also the wrong way round for the framework: hopping describes matter best at high temperature, and it is at low temperature, where band behaviour is cleanest, that RealQM would need it to hold.

5 What remains open: the Fermi surface

The framework has no bands and no E_F, and there are measurements bearing on this directly. They are made on real two-component metals, not on any idealisation, so they cannot be set aside by choosing the object of study more carefully:

- **de Haas–van Alphen oscillations** map the Fermi surface of copper and aluminium directly; its existence and shape are not in doubt;
- the **electronic specific heat** is linear in temperature, with a coefficient fixed by the density of states at E_F;
- the **plasmon dispersion** is positive and measured, while the compressibility term that produces it, $\omega^2 = \omega_p^2 + \frac{3}{5}k^2 v_F^2$, is absent here.

No account of these is currently available in the framework, and constructing one is the substantial open problem — the more so because it survives everything else in this paper. It is not disposed of by the equation of state, since the lattice scale of a simple metal turns out to be electrostatic and to test no quantum treatment at all (§2). It is not disposed of by the conduction repair, since a shared valence territory conducts while still having no band structure and no E_F. And it is not an artefact of any idealisation, being measured on ordinary metals. A set of Fermi-surface phenomena simply has no representation in a theory built from densities on domains, however those domains are constrained.

Remark: the uniform electron gas

It is worth recording what happens if the framework is applied to the *uniform electron gas*, since the calculation is exact and that system is often taken as the reference problem.

Partitioning a gas of density n into non-overlapping cells of size $a = n^{-1/3}$ gives a localisation cost per cell scaling as a^{-2}, hence a kinetic energy density $t(n) = C\,n^{5/3}$ — the Thomas–Fermi form, obtained from the geometry of a partition rather than from Fermi statistics. Linearising (1) about the uniform state and substituting into Poisson's equation gives Yukawa screening with

$k^2 = 18\pi n^{1/3}/5C$, and setting C to the Thomas–Fermi coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces k_{TF} exactly at every density. The *form* extends without adjustment.

The coefficient does not. With the free interface used throughout RealQM, $\psi_i \equiv |\Omega_i|^{-1/2}$ is admissible: it satisfies the normalisation, makes the total density uniform, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Hence $C = 0$ exactly, the screening length is infinite and the plasmon dispersionless. Neither allowing the interfaces to move nor requiring domains to be connected and singly charged alters this.

This should not be read as a test the framework fails. The uniform electron gas is electrons plus a *smeared, inert, non-responding* positive background — a construction of a tradition in which exchange–correlation energies are fitted to that very system. RealQM contains no inert backgrounds anywhere, so asking it to reproduce jellium asks it to reproduce a system whose central ingredient it does not have, and $C = 0$ diagnoses that mismatch. Consistently with this, the pathology disappears as soon as real ions are present: Table 1 is the same framework on the same physics, with the background replaced by the charges matter is actually made of.

One physical case does approach the uniform limit closely: degenerate matter in which the ion lattice is a small perturbation, as in a white dwarf interior, where the measured mass–radius relation depends on precisely the pressure the uniform limit does not supply. That is the one place the result might have observable consequences, and it is untested here.

A note on what was tested numerically

Equation (2), the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. Table 1 is a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].